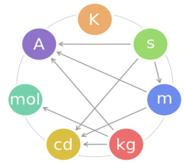
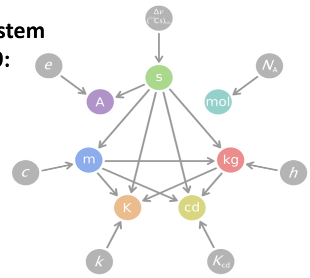
Metrology - *The science of measurements*

- Definition of units
- Accurate methods for making measurements and calibrating measurement instruments
- How to make these widely applicable
- Standardization – SI, International System of Units

Previous SI system:



New SI system from 2019:



Literature:
Read the NIST webpage, it is worth!!!:
https://www.nist.gov/si-redefinition/turning-point-humanity-redefining-worlds-measurement-system
Further material
Nature Physics, The New system of units, **12**, 4 (2016), doi:10.1038/nphys3612
BIPM http://www.bipm.org/en/about-us/
Wikipedia https://en.wikipedia.org/wiki/2019_redefinition_of_SI_base_units

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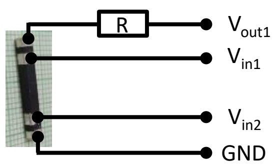
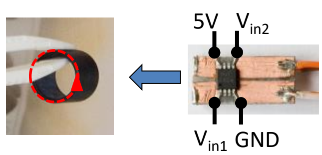
Metrology - Applications

Our example: zero resistance of a superconductor - improved measurement idea yields many orders of magnitude improvement in the measurement error



Four probe resistance measurement
$\Delta\rho \sim 10^{-8}\ \Omega m$



Persistent current measurement with a magnetic field sensorent
$\Delta\rho \sim 10^{-16}\ \Omega m$

Persistent current in the SC solenoid magnet of an NMR spectrometer
$\Delta\rho \sim 10^{-32}\ \Omega m$

Broad range of applications

GPS

Chip production: 80cm diameter with nanometer-scale precision

Robotized manufacturing

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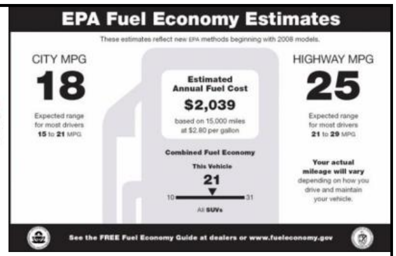
SI – International System of Units



Some history of the SI system:

- In 1875, the process of **standardising units** of measurement began: kg and m standards were established and agreed to be used. (Motivation: 30MPG (Miles per Gallon) = ?l/100km, 1lb/ft^3 (Pound per cubic Foot)= ? kg/m^3)
- **1960 SI system is established**: Internationally agreed definition of standardized units of measurement. Also fixed the method of measuring them. Based on metre (m), kilogram (kg), second (s) and ampere (A) - MKSA. (There are other systems, like CGS. Also, there are other accepted units for everyday use (USA, UK, Canada).
- **7 basic quantities** are fixed and the others are derived from these:

Time	Distance	Mass	Current	Temperature	Amount of substance	luminous intensity
s	m	kg	A	K	mol	cd

- Since 2019, the SI system is based on **fundamentally new principles by fixing natural constants**
- Measurement principles and physical phenomena used for the accurate determination of basic quantities: Atomic clocks, Interferometers, Josephson effect, Quantized Hall effect, Electron pump, Johnson-Nyquist noise, Watt balance, Avogadro project

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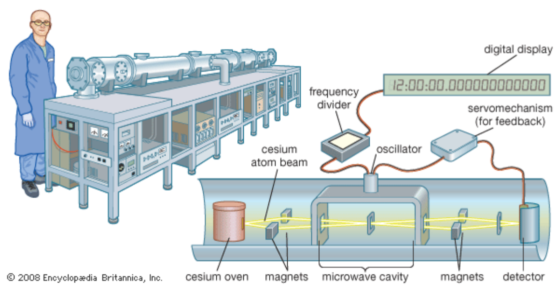
Metrology - Evolution of time measurement

Hourglass (1338). The earliest known hourglass (operated with water instead of sand) dates back to ancient Egypt around the 14th century BC.



Today, atomic clocks are used, these form the basis of almost all other measurements.

The Riefler pendulum clock, which was the primary standard for time interval measurements from 1904 to 1929, is now on display at the NIST museum in Gaithersburg

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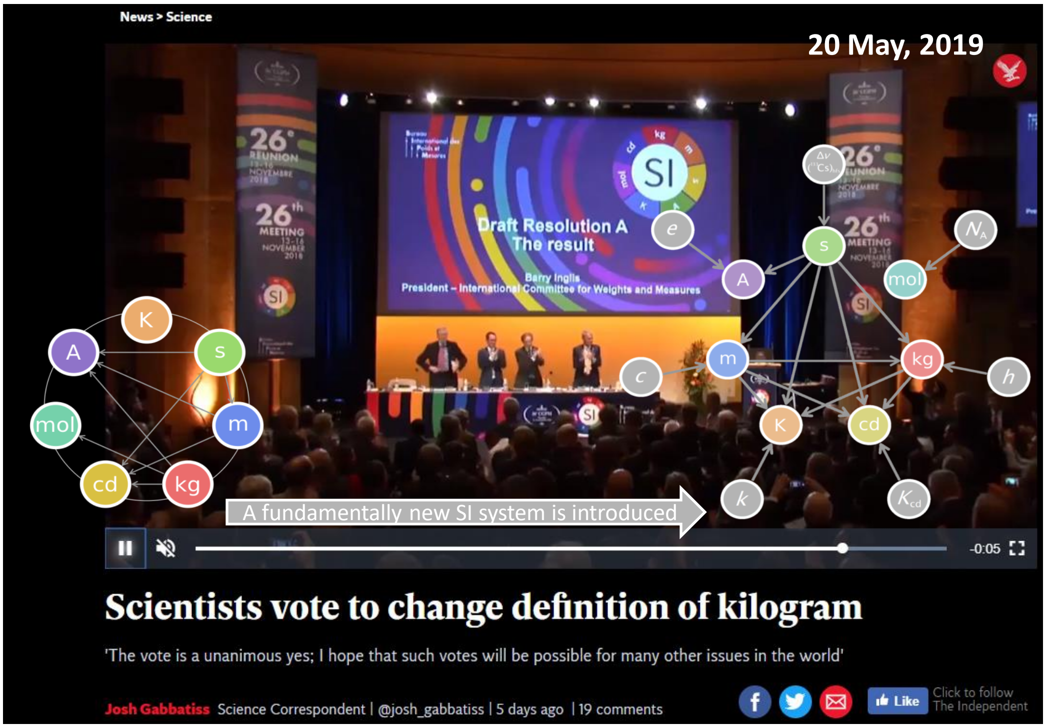

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Metrology - Evolution of distance measurement

(1790) France: The length of a pendulum with a half period of 1sec.

$$T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{l}{g}}.$$

Previous etalon of meter

(1889) Distance between two divisions marked on the etalon rod at the melting point temperature of ice. Accuracy ~10^{-7}

1927 October 6 – The 7th CGPM redefines the metre as the distance, at 0 °C (32 °F), between the axes of the two central lines marked on the prototype bar of platinum-iridium, this bar being subject to one standard atmosphere of pressure and supported on two cylinders of at least 1 cm (0.39 in) diameter, symmetrically placed in the same horizontal plane at a distance of 571 millimetres (22.5 in) from each other.

…

(1983) A metre is the distance that light travels in 1/29979245 seconds.

Reasons for change in definition: new more accurate measurement methods and reproducibility

Forrás: http://en.wikipedia.org/

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Metrology - Evolution of distance measurement

Distance measurement through the speed of light

(1983) The speed of light was fixed at a value of
c = 299 792.458 m/s. This defines the meter by converting it back to seconds through c.

Before that the value of c had to be determined very accurately. For this the $c = \lambda f$ relation of electromagnetic waves was utilized in various approaches:

Electromagnetic standing waves

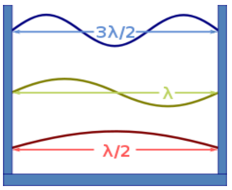
Resonance frequency measurement in a microwave cavity resonator.
Principle: Standing waves are generated in the cavity (see figure and experiment). The resonator size and frequency can be used to calculate c.
(1950) Essen and Gordon-Smith, accuracy ~10^{-7}

Laser interferometer

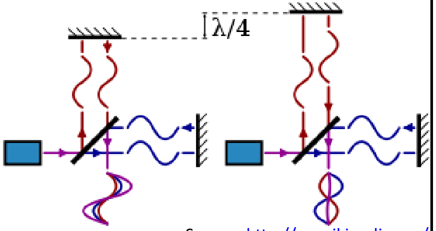
Principle: - Sending laser light of a known frequency through two paths, the two beams either strengthen or cancel each other depending on the path difference.
(1972) NIST determination of c with a laser interferometer
Accuracy: 3.5×10^{-9}

Experiment: standing waves in a microwave oven

Source: http://www.fizikaiszemle.hu/archivum/fsz0503/hartlein0503.html



Operating principle of a Michelson interferometer



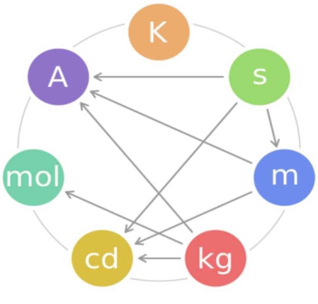
Source: http://en.wikipedia.org/

SI – Definitions of basic units before 2019

Definitions in the previous SI (before 2019)

s	Part 1/9192631770 of the hyperfine transition period of caesium 133.
K	Part 1/273.16 of the triple point temperature of water.
m	Converted back to seconds (s) based on the c=299 792.458 m/s speed of light.
mol	Number of atoms in 12g of C^{12}.
A	The force per unit length of 1 m between two current-carrying conductors (1A) 1 m apart is 2x10-7 N .
kg	Mass of the international Pt mass reference.
cd	The luminous intensity of a monochromatic (f=540×1012 Hz) source emitting 1/683W per unit solid angle.

Relations in the previous SI



Of course, there are further derived units of measurement, example: power
$W=J/s=kg\ m^2\ s^{-3}$

New SI

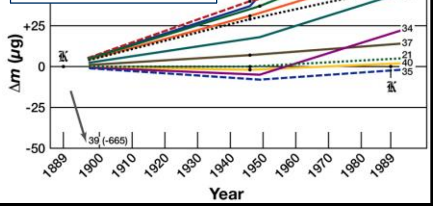
Max Plank (1900) *Ann.Physik 1, 69-122 (1900)*
*"...with the help of **fundamental constants** we have the possibility of establishing **units of length, time, mass, and temperature**, which necessarily retain their significance for all cultures, even unearthly and nonhuman ones."*

New-SI: redefinition of 4 out of 7 basic units

- 2011: proposal of the *General Conference on Weights and Measures.* Accepted in 2019.
- Natural constant values are defined, and their consequences are the units of measurement. Concerned natural constants: c, e, h, k, N_A
- This philosophy was already partially included in the previous SI system through the [s] x c = [m] relation defining the meter (1983).
- Some problems with the previous SI system:
  - Kilogram: linked to the reference mass of the Pt standard. If its mass changes over the years (e.g. due to oxidation), then each measured mass value varies globally (>75µg variation over the years).
  - Kelvin: (triple point of water) Purity, salts, other contaminations, O and H isotopes change the value of the triple point. Furthermore, it defines only a single temperature point.
  - Ampere: a very unnatural definition, especially if we could count the charges one by one
- Goal: more accurate and simpler definitions
  target accuracy ~10^{-8}

Pt cylinder: international prototype kilogram (IPK)

Mass drift of national prototype kilograms and two IPK sister prototype kilograms relative to the IPK (Wikipedia)



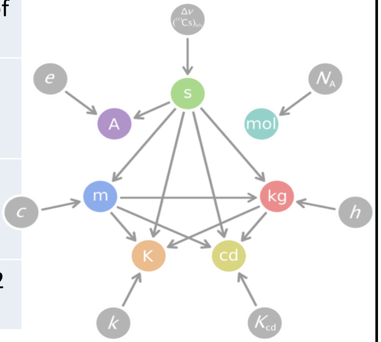
New SI system

New definitions:

s	The frequency of the ground state hyperfine splitting of the caesium-133 atom, $\Delta\nu$ (^{133}Cs) is fixed. Time (s) is measured compared to that frequency (t=1/ν).
K	Temperature (K) is converted back to other units (s, m, kg) by fixing the Boltzmann constant. (E~ *k*T, [E]=kg*m²/s²)
m	Distance (m) is converted to time (s) by fixing the *c* speed of light (s=c*t)
mol	The amount of substance (mol) is converted to number of elementary entities by fixing the Avogadro constant, N_A.
A	Current (A) measurement is converted to time (s) measurement through fixing the *e* elementary charge (Q= I*t).
kg	Mass (kg) meas. is converted to time and distance measurement through fixing the h Planck constant (E=h*ν, [E]=kg*m²/s², [ν]=1/s)
cd	The luminous intensity of a monochromatic (f=540×1012 Hz) source emitting 1/683W per unit solid angle.

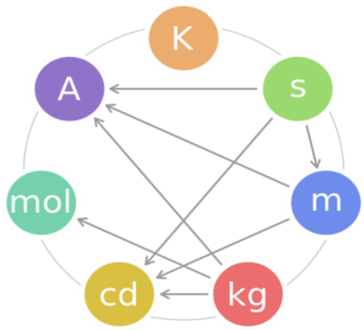
Fixed natural constants:

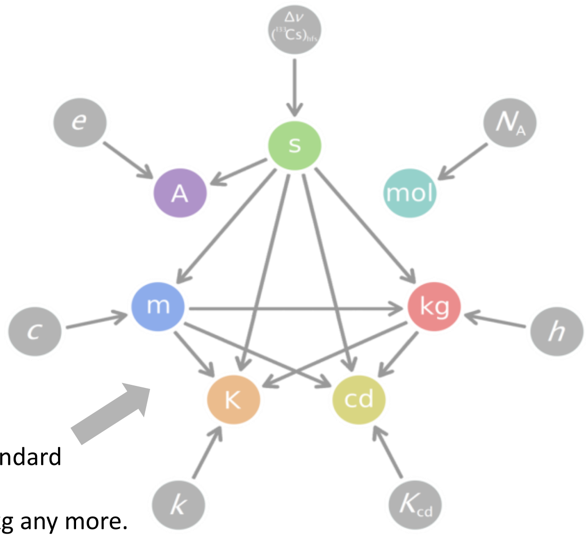
$\Delta\nu(^{133}\mathrm{Cs})_{hfs} = 9192631770\mathrm{s}^{-1}$
$c = 299792458\mathrm{s}^{-1}\cdot\mathrm{m}$
$h = 6.62606\mathrm{X}\times10^{-34}\mathrm{s}^{-1}\cdot\mathrm{m}^2\cdot\mathrm{kg}$
$e = 1.60217\mathrm{X}\times10^{-19}\mathrm{s}\cdot\mathrm{A}$
$k = 1.38065\mathrm{X}\times10^{-23}\mathrm{s}^{-2}\cdot\mathrm{m}^2\cdot\mathrm{kg}\cdot\mathrm{K}^{-1}$
$N_A = 6.02214\mathrm{X}\times10^{23}\mathrm{mol}^{-1}$
$K_{cd} = 683\ \mathrm{s}^3\cdot\mathrm{m}^{-2}\cdot\mathrm{kg}^{-1}\cdot\mathrm{cd}\cdot\mathrm{sr}$



New and old SI system

Relations of the quantities





Changes in the relations:

- -No more etalons, like the Pt mass standard
- -Kilogram depends on m and s.
- -Ampere does not depend on m and kg any more.
- -Mol does not depend on kg any more.
- -Kelvin depends on m, kg and s.

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Measurement methods and related physical phenomena

	Definition	Measurement principles, standards and related quantum phenomena
s	Fixing the hyperfine splitting of the Caesium 133 atom, $\Delta\nu(^{133}Cs)$.	^{133}Cs atomic clocks
K	Fixing the Boltzmann constant (k).	Speed of sound in noble gases, Johnson-Nyquist noise
m	Fixing the speed of light (c).	Laser light interference
mol	Fixing the Avogadro number(N_A).	-
A	Fixing the elementary charge (e).	Josephson effect + Quantized Hall effect (or electron pump?)
kg	Fixing the Planck constant (h).	Watt balance, Avogadro-project
cd	The luminous intensity of a monochromatic (f=540×1012 Hz) source emitting 1/683W per unit solid angle.	…

Already discussed before

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